

Career Profile — Tarvin Kaur Arora

Last updated: 2026-07-29. This is a long-form, honest source-of-truth — not a resume. Resumes are tailored outputs derived from this file.

1. Identity

- **Name:** Tarvin Kaur Arora
- **Location:** Vancouver, BC, Canada (studying at UBC)
- **Email:** tarvinkaur@outlook.com
- **Phone:** +1 236 308 4654
- **LinkedIn:** www.linkedin.com/in/tarvin-kaur-arora/
- **GitHub:** github.com/tarvinarora

2. Headline & target roles

- **Current headline:** Statistics undergrad (UBC, grad May 2027) working across data science, analytics, and applied statistics — with hands-on experience building components of a deployed internal AI product.
- **Target role title(s):** Data Scientist (new-grad), Data / Business Analyst, ML / Data Science Intern, Quantitative / Statistical Analyst
- **Track:** IC
- **Target geography:** Canada
- **Timing:** New-grad full-time (available after May 2027 graduation)
- **Career switcher?** No

3. Summary

Statistics undergraduate at UBC (graduating May 2027) with a strong applied-statistics and data-science foundation. Interned at Mercer (Marsh McLennan) as the sole data science intern, where I built the OCR extraction and knowledge base behind a deployed internal RAG-based AI product, plus a Power BI benchmarking dashboard and Excel automation. Comfortable across the analytics stack — Python, R, SQL/MongoDB, Power BI — with genuine depth in experimental design and statistical inference (factorial experiments, ANOVA, hypothesis testing). I care about honest, readable data communication and am currently building out agentic-workflow / API projects.

4. Experience snapshot

Role	Company	Dates	Location	Team size
Undergraduate Teaching Assistant	University of British Columbia	Sep 2026 – Present	Vancouver, BC	Teaching team (labs)
Data Science Intern	Marsh McLennan (Mercer Consulting Pvt. Ltd.)	Jun 2025 – Aug 2025	(confirm — Mercer office)	Sole intern on AI product team

Role	Company	Dates	Location	Team size
STEM & Language Tutor	Varsity Tutors	Jan 2024 – Present	Remote	Individual

5. Experience deep-dive

Data Science Intern @ Marsh McLennan (Mercer Consulting Pvt. Ltd.) (Jun 2025 – Aug 2025)

- **Team:** AI product team building an internal, RAG-based AI product. I was the **only intern**. Reported to the Product Manager and the AI Head, Aditya Gupta.
- **Stack:** Python; OCR (PyTesseract, pdf2image, FuzzyWuzzy); RAG (leveraged an existing pipeline); SharePoint integration; backend LLM API; Power BI (Power Query, DAX); Excel macros.
- **Scope:** Internal AI product, **deployed** and integrated with SharePoint on a backend LLM API. 1,210 fields extracted for the InfoSec assistant.
- **Key projects:**
 - **InfoSec query chat assistant** — Built OCR-based field extraction from client documents (end-to-end, mine) and built the **knowledge base** that fed an existing RAG pipeline (structured knowledge documents, source-constrained to approved client docs). Enabled automated Q&A retrieval, reducing client-legal query turnaround from ~14 days to near-instant across 1,210 extracted fields. (*Turnaround figure is a team estimate — defensible, approximate.*)
 - **Power BI compensation benchmarking dashboard** — 12+ KPIs using Power Query transformations, DAX measures, and dynamic visuals for remuneration analysis, used in client presentations. Made deliberate viz-design calls: dropped percentages that highlighted stats that weren't actually impactful, added market benchmarking and filters for genuine insight, and used graphics/icons and negative space to improve readability.
 - **Excel macro automation** — Workflow-based automation for project tracking; synchronized milestones and status updates across teams, reducing manual update effort by ~60–70%. (*Team estimate.*)
 - **UAT testing** — End-to-end user-acceptance testing for chatbot features (retrieval quality, edge cases, user flows, response consistency); documented defects, user guides, FAQs, and release notes for internal deployment.
- **Metrics / outcomes:** 1,210 fields extracted; ~14 days → near-instant retrieval (team estimate); ~60–70% manual-effort reduction (team estimate); 12+ KPIs in the dashboard.
- **Cross-functional:** Worked directly with the Product Manager and AI Head; product served internal/client-legal Q&A needs.
- **Ownership note:** OCR extraction, the knowledge base, the Power BI dashboard, and the Excel automation were all mine. I **leveraged** the RAG pipeline; I did not build it (see Section 19).
- **Adjacent understanding gained:** Industry-standpoint understanding of cloud architecture, from exposure to how the RAG product was deployed (SharePoint + LLM API + cloud). Exposure-level, not hands-on infra configuration.

Undergraduate Teaching Assistant @ University of British Columbia (Sep 2026 – Present)

- **Team:** Teaching team for an introductory computational-thinking course.

- **What I do:** Lead two in-person weekly lab sessions, guiding students through hands-on exercises in data representation, algorithms, and foundational programming. Grade student work with constructive feedback and contribute to lab and course-project materials.

STEM & Language Tutor @ Varsity Tutors (Jan 2024 – Present)

- **Scope:** 50+ students tutored in statistics, mathematics, and computer science.
- **What I do:** Strengthen foundations in probability, data analysis, programming logic, and quantitative problem-solving. Support college-level analytics/statistics coursework (EDA, multivariate linear regression, hypothesis testing, residual analysis, model evaluation, data visualization, and debugging). Maintain a 5-star rating through clear, adaptive teaching.

6. Education

Degree	Institution	Year	Notes
Bachelor of Science in Statistics	University of British Columbia	Expected May 2027	GPA omitted from resumes (not a headline strength)

7. Certifications

Certification	Issuer	Year	Notes
DELFDALF B1 (French)	France Éducation internationale	—	Confirm year
IELTS — 7.5	British Council / IELTS	—	Confirm year

8. Technical skills

- **Languages:** Python, R, Java
- **Databases:** SQL, MongoDB
- **Data science & statistics:** R (tidyverse, ggplot2, dplyr, glmnet), Python (pandas, NumPy, matplotlib, seaborn, scikit-learn, PyTorch); experimental design, factorial experiments, ANOVA, hypothesis testing, regression, PCA, clustering (K-means, SOM)
- **Visualization / BI:** Power BI (Power Query, DAX), Tableau, Altair, matplotlib, seaborn, ggplot2, Excel
- **Tooling:** Git/GitHub, VS Code
- **OCR & automation:** PyTesseract, pdf2image, FuzzyWuzzy, Excel macros
- **Emerging (building depth):** Agentic workflows / API integration, RAG (knowledge-base side), industry cloud-architecture understanding

9. Domain expertise

- **Applied statistics & experimental design** — the strongest area: factorial/experimental design, ANOVA, hypothesis testing, statistical validation with residual diagnostics.
- **Data science & analytics for business** — dashboards, benchmarking, and honest data communication for stakeholder/client audiences.
- **AI product exposure** — hands-on with the ingestion/knowledge-base side of a deployed RAG product, plus adjacent understanding of cloud deployment.

10. Leadership & people

- **Teaching:** Lead weekly lab sections as a UBC TA; contribute to course materials.
- **Mentorship:** 50+ students tutored across STEM.
- **Initiative:** Volunteered with two UBC professors on course development over a summer (see Section 14).

11. Public artifacts

- **GitHub:** github.com/tarvinarora
- *(No talks / publications / OSS yet — potential build area.)*

12. Languages spoken

Language	Proficiency
English	Fluent (IELTS 7.5)
French	Intermediate (DELF/DALF B1)
Punjabi	Fluent
Hindi	Fluent

(Often dropped from Canadian/US tailored resumes; French may be worth keeping for bilingual roles.)

13. Awards & recognition

(None captured yet.)

14. Volunteer / community

- **Course development with two UBC professors (summer):**
 - **DSCI-320 restructuring** — thorough analytics on AI usage by students to inform course redesign.
 - **CPSC 436C (Cloud Computing)** — built simulator exercises for students.

15. Interests

(Omitted by preference.)

16. Visa / work authorization

International student in Canada. Will require a Post-Graduation Work Permit (PGWP) for post-graduation employment. Relevant to Canadian new-grad postings that ask about work authorization.

17. Achievements bank

- Built the OCR-based field extraction (PyTesseract, pdf2image, FuzzyWuzzy) for a deployed internal InfoSec AI assistant — 1,210 fields extracted from client documents.

- Built the knowledge base feeding an existing RAG pipeline, source-constrained to approved client documents, enabling automated Q&A retrieval that cut client-legal query turnaround from ~14 days to near-instant (team estimate).
- Built a Power BI compensation benchmarking dashboard (12+ KPIs, Power Query, DAX, dynamic visuals) used in client presentations.
- Made deliberate data-viz calls: cut vanity percentages, added market benchmarking and filters for real insight, used icons/negative space for readability.
- Built Excel macro automation for cross-team project tracking, cutting manual update effort ~60–70% (team estimate).
- Ran end-to-end UAT for chatbot features and authored user guides, FAQs, defect logs, and release notes for an internal deployment.
- Designed relational SQL and MongoDB schemas for a streaming-analytics database (Netflix/YouTube), architected entity relationships, and demonstrated when SQL vs. NoSQL should apply — team project, **presented to faculty**.
- Ran a factorial experiment on neural-network performance: 20+ preprocessing/architecture/capacity combinations across 5,000 simulated points; MLP/CNN/LSTM in PyTorch; ANOVA-based inference across Latin-square, fractional, and full factorial designs, validated with residual diagnostics and Q-Q plots.
- Characterized seismic signals (real earthquake data from multiple Mexican stations) using FFT spectral features, Self-Organizing Maps, and K-means; evaluated cluster stability with cosine similarity (0.85–0.97) and PCA.
- Tutored 50+ STEM students with a sustained 5-star rating.
- Volunteered on UBC course development (DSCI-320 redesign via student-AI-usage analytics; CPSC 436C simulator exercises).

18. Known gaps → build / revise targets

(Framed as things to strengthen or build, not disclaimers. Prep before interviews; don't undercut the resume.)

- **Agentic workflows / API project** — in progress, not yet built. Target: build this out (likely to replace the seismic project as the third featured project).
- **Seismic clustering project** — candidate plans to swap it out for the agentic-workflows/API project once ready; keep here until the replacement is done.
- **Lower-confidence tools** — where depth is thinner, the plan is to prepare/refresh before interviews and, where useful, build a small project to solidify the skill rather than omit it.
- **Public artifacts** — no talks/OSS/publications yet; a GitHub-hosted agentic-workflow project would fill this.

19. Pre-built framings

- **Topic:** RAG / AI product ownership **Approved framing:** "Built the OCR extraction and knowledge base feeding an existing RAG pipeline." **Never:** "Built the RAG pipeline." **Why:** Leveraged the existing pipeline; ownership was the ingestion/knowledge-base side. Defensible and specific.
- **Topic:** Cloud architecture **Approved framing:** "Developed an industry-standpoint understanding of cloud architecture through exposure to a deployed RAG product." **Never:** "Configured / owned cloud infrastructure." **Why:** Exposure-level understanding, not hands-on infra work.

- **Topic:** Impact metrics (14 days → near-instant; 60–70% effort reduction) **Approved framing:** Present as approximate team estimates ("reduced ... by roughly 60–70%", "from ~14 days to near-instant"). **Why:** Team estimates — provable and defensible, but approximate. Don't present as precisely measured personal metrics.